Introduction to Robotics, Homework 1

The AR-60TM arm

Forward and inverse kinematics, differential kinematics, singularities

MIPT

Center for Cognitive Modeling

What this homework is

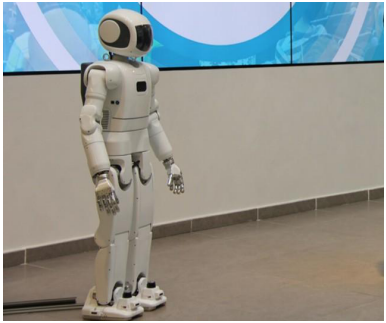
One arm, six tasks, eighteen hours

AR-601M is a 41-DOF humanoid robot. This homework treats one of its two arms in isolation: a 6-DOF serial chain, shoulder to tool, with a spherical wrist.

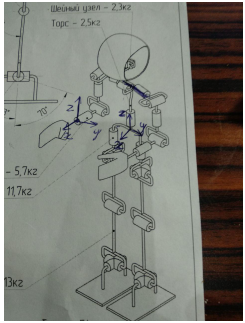
- **Forward** kinematics: joint angles → tool pose.
- **Inverse** kinematics, analytical and numerical: tool pose → joint angles.
- **Differential kinematics** and **singularities**.

About 18 hours of work covering weeks 1–3. This deck is a pointer; the full statement, with every derivation, requirement and hint, is the handout linked on the last frame.

The mechanism: AR-601M



(a) The AR-601M arm.



(b) The blueprint, annotated in blue.

Figure: The robot, and the blueprint it is modelled from; lengths are not legible on the blueprint, see handout §2.

The six tasks

- **1.** Kinematic model
- **2.** FK: angles → arm and pose
- **3.** Analytical IK: pose → all feasible branches
- **4.** Numerical IK: solve from multiple seeds
- **5.** Differential kinematics
- **6.** Singularity and solution-count maps

Full requirements and evidence for every task are in handout §4.

Required visualizations

1. **FK:** enter six joint angles; draw the manipulator and print the resulting homogeneous matrix T.
2. **Analytical IK:** enter a pose from FK or a position and ZYX Euler orientation; draw *every distinct feasible* configuration and print each complete six-angle vector.
3. **Singularities:** plot singularity information and a fixed-orientation workspace slice. Use a continuous color scale for the number of feasible IK configurations, from zero to the maximum attained in this case.

Mark unresolved targets and continuous families separately from finite counts.

What you hand in

The packaging contract

- A `pixi.toml` or a `Dockerfile` at the repository root.
- One command reproduces the report's numbers, example images and maps from a clean checkout.
- Three operations: `FK(q, robot)`, `IK(T, robot)`, `DifferentialKinematics(q, robot)` — any language.

Packaging is what is graded, not the language used to write it.

The full statement

Everything above is a pointer. The handout has every derivation, every requirement and every hint.

Read handout §3 and §4 before writing any code.

hw01_handout.pdf on GitHub



The handout

Deserve “A” grade!

– Oleg Bulichev

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